Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/09/2009
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Effect of Spaceflight on Microbial Gene Expression and Virulence		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	06/01/2006	**End Date:**	01/31/2010
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-115 , STS-123 NOTE: Received NCE to 1/31/2010, per A. Chu/ARC (9/08) NOTE: End date changed to 1/31/2009 per A. Chu/ARC (6/08)		
Key Personnel Changes/Previous PI:	PI moved to Arizona State University from Tulane University mid-2006.		
COI Name (Institution):	Schurr, Michael (University of Colorado Health Sciences Center) Buchanan, Kent (Oklahoma City University)		
Grant/Contract No.:	NNA06CB26A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Human presence in space, whether permanent or transient, will be accompanied by the presence of microbes. The changes that occur to microorganisms during space flight are not fully understood, and the potential of a negative impact upon health, safety, and performance of crewmembers requires further investigation. Space flight studies have demonstrated changes in bacterial and fungal growth, morphology, and metabolism, and ground-based studies have indicated changes in bacterial virulence characteristics and protein production during growth in modeled microgravity, a laboratory environment that stimulates some aspects of spaceflight. Thus, during long-duration space flight, the virulence of pathogenic microbes may be enhanced and normally harmless commensal or environmental microbes may become pathogenic. This effect would be intensified by several factors, including: 1) the increased duration of missions, 2) a greater microbial diversity resulting from the use of regenerative life support systems, and 3) a potentially compromised human immune system during flight. Investigation into the effects of space flight on microbial characteristics is difficult as flight experiments are expensive and opportunities are rare. Therefore, ground-based analog systems have been developed for cellular and microbial testing; the most popular system being NASA designed rotating wall vessel (RWV) bioreactors, which simulate many of the mechanistic effects of microgravity on biological systems. The objective of this research is to compare microbial gene expression and virulence attributes in microorganisms during extended space flight to the same organisms cultured under normal gravity and in the modeled microgravity of the RWV. The latter comparison is essential to verify the applicability of the RWV bioreactors to simulate in-flight conditions and to use this system to access the probability of in-flight infectious disease. This proposal is to fly two copies of the automated single locker MOBIAS (Multiple Orbital Bioreactor with Instrumentation and Automated Sampling) culture unit, to allow simultaneous analysis in multiple replicates of four difference bacterial and fungi. This study develops a foundation from which future in-flight research can be based and which may also hold commercial promise for translation to the clinical bedside. The objective of this research is to determine changes in microbial gene expression and virulence attributes during extended spaceflight. These results will be compared to those obtained when the same experiments are performed with the identical microbes cultured under ground-based conditions of modeled microgravity (MMG) in NASA-designed bioreactors. This comparison is vital in determining the value of the NASA-designed bioreactors in suspension culture of microbes under MMG for use in: (a) simulating the probability of in-flight infectious disease; and, (b) developing commercial potential for the bioreactors for translation to the clinical bedside. Four model microbes, two bacterial and two fungal, are proposed for use in these studies: a). the obligate pathogen Salmonella typhimurium, b). the opportunistic pathogen, Pseudomonas aeruginosa, c). the commensal organism and opportunistic pathogen, Candida albicans; and, d). the classic model eukaryotic microbe, Saccharomyces cerevisiae. These organisms were shown to be gravity sensitive in earlier flight experiments and/or ground-based simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will develop a foundation from which future ground-based research can be based and also holds commercial promise for translation to the clinical bedside. These investigations will be the first of their kind to investigate comprehensive changes in microbial gene expression which may occur during extended spaceflight and their comparison to the expression profiles of the same microbes grown as ground-based suspension cultures under MMG in the NASA-designed bioreactors. The latter comparison is essential to verify the applicability of the MMG bioreactors to simulate in-flight conditions and to use this system to assess the probability of in-flight infectious disease. However, these studies will also provide a solid foundation for testing the commercial applicability of ground-based microbial suspension culture under MMG for the development of vaccines and other novel countermeasures, which are not achievable by any other ground-based means, for the treatment and prevention of infectious diseases occurring on Earth. For example, the PI has previously demonstrated that when cultivated under MMG, the enteric pathogen, S. typhimurium, is more potent in terms of its disease-causing potential. Expanding this work to include other important pathogens could provide significant insight about the microbial response to MMG, and could provide clues for the functioning of known virulence systems, or the identification of novel, uncharacterized microbial virulence strategies, which could lead directly to the development of countermeasure for infectious human disease. Importantly, the knowledge gained from these flight and ground-based studies has identified the existence of a previously unknown mechanism that affects microbial virulence. By targeting and understanding this new mechanism, novel therapeutics and vaccines may be developed against infectious disease.
	We have flown two Shuttle spaceflight experiments as part of this funded grant – first on STS-115 (MICROBE) and then on STS-123 (MDRV). The purpose of the MICROBE experiment was to examine the global effects of spaceflight on microbial gene expression and virulence attributes. This experiment was the first study to examine the effect of spaceflight on the virulence of a pathogen, and the first to obtain the entire gene expression response of a bacterium to spaceflight. In this experiment, we performed microarray-mediated and proteomic global gene expression profiling of the microbial pathogens S. typhimurium, P. aeruginosa, and C. albicans in response to spaceflight as compared to identical synchronous ground controls and determined the effect of spaceflight on the virulence potential of these microorganisms using murine models. The model microorganisms were selected as they have been isolated from preflight or in-flight monitoring, represent different degrees of pathogenic behavior, are well characterized, and have sequenced genomes with available microarrays. In particular, extensive studies of S. typhimurium by the PI's lab using ground-based spaceflight analog systems demonstrated important changes in the genotypic, phenotypic, and virulence characteristics of this pathogen resulting from exposure to a flight-like environment (i.e. modeled microgravity). Published results from MICROBE showed that Salmonella grown in rich LB media in spaceflight displayed increased virulence, biofilm-like formation, and global alterations in global gene expression, as compared to synchronous ground controls. A central role was identified for the conserved, small regulatory RNA-binding protein Hfq in regulating key aspects of Salmonella's spaceflight microgravity response. In addition, many genes encoding ion response pathways showed altered expression during spaceflight, reinforcing our earlier results observed in the spaceflight-analogue RWV bioreactor culture of S. typhimurium. These results led us to hypothesize that media ion concentrations could be manipulated to prevent/turn off the enhanced Salmonella virulence imparted during spaceflight. This hypothesis was tested in the MDRV experiment – which was flown on mission STS-123. The MDRV experiment also provided us with the opportunity to independently validate our virulence and gene expression results obtained from MICROBE on an independent Shuttle mission. Consistent with our hypothesis, we found that spaceflight-induced increases in Salmonella virulence are regulated by media ion composition, and that phosphate ion is sufficient to alter related pathogenesis responses in a spaceflight analogue model. Specifically, the increased S. typhimurium virulence observed with cultures grown in spaceflight in rich

Task Progress:	LB medium was not exhibited with cultures grown in M9 minimal medium. Based upon our quantified differences in ion concentrations between LB and M9 media, we supplemented LB medium with inorganic ions to the same levels as those found in M9 medium and tested the effect on virulence during spaceflight. We found that this ion supplementation was sufficient to prevent the enhanced Salmonella virulence imparted during flight. Subsequent testing in ground-based spaceflight analogue culture conditions indicted that the altered acid tolerance exhibited by Salmonella during culture in LB alone was prevented with the addition of inorganic phosphate. These results demonstrate a direct correlation between phosphate ion concentration and the phenotypic response of Salmonella to the environment of spaceflight analogue culture. We also showed that spaceflight uniquely exacerbates the Salmonella virulence effect in an ion-dependent fashion. These findings indicate that virulence in Salmonella can be influenced by the environmental stimulus of spaceflight, and that the response to this stimulus can be manipulated to improve astronaut health measures and exploited to better understand microbial pathogenesis and develop innovative therapeutics. Finally, despite the multiple phenotypic differences in response to spaceflight between the two media (LB and M9), we found that several common genes and gene families were altered in expression in both media during spaceflight culture. We reported that as with spaceflight growth in rich LB media, Salmonella grown in minimal M9 media during flight displayed differential expression of many genes, including those associated with either the regulation of, or regulation by, the Hfq protein and small regulatory RNAs. Identification of these genes, whose expression is commonly regulated by the low fluid shear environment of spaceflight, provides key targets whose expression can be manipulated to control microbial responses, including potential use for development of vaccines and therapeutics. The results of these studies with Salmonella, and future publications with the other pathogens flown, will provide important new insight for microbial risk assessment for both long and short duration spaceflight missions. In addition, by understanding the changes that pathogenic microorganisms undergo in the space environment, these studies may lead to the development of novel vaccines and countermeasures for the treatment and prevention of infectious diseases occurring during spaceflight and on Earth.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Wilson JW, Ott CM, Quick L, Davis R, zu Bentrup KH, Crabbé A, Richter E, Sarker S, Barrila J, Porwollik S, Cheng P, McClelland M, Tsaprailis G, Radabaugh T, Hunt A, Shah M, Nelman-Gonzalez M, Hing S, Parra M, Dumars P, Norwood K, Bober R, Devich J, Ruggles A, CdeBaca A, Narayan S, Benjamin J, Goulart C, Rupert M, Catella L, Schurr MJ, Buchanan K, Morici L, McCracken J, Porter MD, Pierson DL, Smith SM, Mergeay M, Leys N, Stefanyshyn-Piper HM, Gorie D, Nickerson CA. "Media ion composition controls regulatory and virulence response of Salmonella in spaceflight." PLoS ONE. 2008;3(12):e3923. PMID: 19079590 , Dec-2008